Daffodil International University

Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25010
Project Title	IoT-based Smart Home System
Semester	Spring 2025
Students (ID – Name)	221-50-083 – Md. Nazmul Hosen Bondon 221-50-080 – Rimpe Deb 221-50-104 – Rabeya Akther Neha
Supervisor	Prof. Dr. A.K.M. Fazlul Haque, Professor, Department of ICE

Synopsis

Problem	Household appliances are usually operated manually, which wastes energy and gives no remote monitoring or early warning of hazards such as gas leakage or fire.
Objective	To design and implement an IoT-based system that monitors home conditions and allows users to control appliances remotely and automatically.
Proposed Method	A microcontroller (e.g. ESP32) with temperature, motion, light and gas sensors and relay modules; sensor data sent over Wi-Fi to a cloud platform; a mobile/web dashboard for monitoring, control and alerts; rule-based automation for lighting and appliances.
Expected Outcome	A working prototype with remote appliance control, real-time monitoring, safety alerts and reduced energy consumption.

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	
Designation	
Organization	
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____

Daffodil International University

Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25006
Project Title	MediAI: A Comprehensive AI System for Medical Diagnosis and Treatment Suggestions
Semester	Spring 2025
Students (ID – Name)	221-50-094 – Aishi Biswas 221-50-099 – Md. Al Mustakim 221-50-097 – Fujilatunnesa Chowdhury
Supervisor	Md. Azizul Hakim, Lecturer, Department of ICE

Synopsis

Problem	Online health information is often unreliable, and patients with low health literacy misunderstand prescriptions and make unsafe self-guided decisions.
Objective	To develop an AI-powered medical assistant that gives safe, explainable health information, explains prescriptions in plain language and supports diet planning, without replacing a doctor.
Proposed Method	Retrieval-Augmented Generation (RAG) on verified medical textbooks, clinical guidelines and drug references; OCR and medical entity extraction for prescriptions; a diet-planning module; a safety layer with confidence filtering and medical disclaimers.
Expected Outcome	A chatbot that answers medical queries with low hallucination, explains prescriptions accurately and improves patient understanding (final report: "AI-Powered Medical Assistant (Medi Chatbot)", December 2025).

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	
Designation	
Organization	
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____

Daffodil International University

Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25004
Project Title	Development of IoT-based Smart Prepaid Energy Meter
Semester	Spring 2025
Students (ID – Name)	221-50-100 – Abdiasis Salad Ahmed 221-50-101 – Abdihakim Mohamud Ahmed
Supervisor	Engr. Md. Zahirul Islam, Assistant Professor, Department of ICE

Synopsis

Problem	Conventional energy meters need manual reading and recharge, causing billing errors, delays and poor visibility of consumption for users.
Objective	To build a smart prepaid energy meter that measures consumption in real time, deducts balance automatically and lets users monitor and recharge online.
Proposed Method	Current and voltage sensing with a microcontroller; energy calculation and prepaid balance logic; relay-based automatic cut-off on zero balance; IoT connection to a cloud database and mobile/web app for recharge, usage history and low-balance alerts.
Expected Outcome	A prototype prepaid meter with real-time consumption monitoring, online recharge, automatic cut-off and alerts.

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	
Designation	
Organization	
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____

Daffodil International University

Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25003
Project Title	Developing an AI Integrated Smart Mentoring System (SMS) with Recommendation System Architecture
Semester	Spring 2025
Students (ID – Name)	221-50-076 – Ashraful Islam Ratul 221-50-077 – Md. Sabbir Ahmed 221-50-109 – Md. Piash Islam
Supervisor	Dipto Biswas, Lecturer, Department of ICE

Synopsis

Problem	Students often lack personalized guidance on courses, skills and career paths, and manual mentoring cannot scale to large batches.
Objective	To develop an AI-integrated mentoring platform that recommends learning resources, skills and mentors to each student based on their profile and progress.
Proposed Method	Student profile and performance data; recommendation engine combining content-based and collaborative filtering; web application for students and mentors; feedback loop to improve recommendations.
Expected Outcome	A working mentoring system that gives personalized recommendations and connects students with suitable mentors.

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	
Designation	
Organization	
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____

Daffodil International University

Department of Information and Communication Engineering

Capstone Project Synopsis – Industry Expert Review

Serial No.	SP25002
Project Title	Implementing Smart Irrigation System: Real-Time Moisture Dependent Decision-making for Irrigation through AI-based Weather Station
Semester	Spring 2025
Students (ID – Name)	221-50-081 – Methon Kumar Biswas 221-50-082 – Sethi Das 221-50-107 – Kazi Rahima Ahmed
Supervisor	Dipto Biswas, Lecturer, Department of ICE

Synopsis

Problem	Fixed-schedule irrigation wastes water and energy and does not respond to actual soil moisture or expected rainfall.
Objective	To design an irrigation system that decides when and how much to irrigate from real-time soil moisture and weather station data.
Proposed Method	Soil moisture, temperature and humidity sensors with a microcontroller; a weather station and weather forecast data; an AI/ML model for irrigation decisions; automatic pump control and a dashboard for farmers.
Expected Outcome	A prototype that irrigates only when needed, reducing water use while keeping soil moisture at the required level.

Industry Expert Review

Industry relevance	☐ High ☐ Moderate ☐ Low
Comments on the topic	
Suggested modifications (scope, tools, standards, data, testing)	
Recommendation	☐ Approved as proposed ☐ Approved with modifications ☐ Not recommended
Name	
Designation	
Organization	
Signature	
Date of review	

Supervisor: _____ Head, Department of ICE: _____